Replacing Newton’s 1st Law with Two New Laws

by Sven Nilsen, 2026

In this paper I argue that a body does not remains at rest, or in motion at a constant speed in a straight line, unless it is acted upon by a force. More precisely, observers see typical histories of a body when they measure outcomes in experiments. These typical histories are depending on the distribution of observers like themselves across the whole universe, most which are unobservable. Typical histories of a body will be a selection effect acting on random events. As a result, Newton’s 1st law might be replaced by two new laws, one that a body moves by a series of random events and two that observers see typical histories of a body in experiments. Newton’s 1st law emerges as a statistical phenomena that results from the universe expanding rapidly.

Newton's laws of motion are three physical laws that describe the relationship between the motion of an object and the forces acting on it. These laws, which provide the basis for Newtonian mechanics, can be paraphrased as follows:

1. A body remains at rest, or in motion at a constant speed in a straight line, unless it is acted upon by a force.
2. At any instant of time, the net force on a body is equal to the body's acceleration multiplied by its mass or, equivalently, the rate at which the body's momentum is changing with time.
3. If two bodies exert forces on each other, these forces have the same magnitude but opposite directions.

In modern physics, Newton’s laws of motion are replaced by quantum physics or relativity theory. Therefore, Newton’s laws of motion are not treated as fundamental laws, but as approximations. This can make it easy to regard some underlying theory as the “more true” picture of what Newton’s laws mean and scientists ought to figure out what this picture should be. However, it might be useful to start with Newton’s laws of motion and change them to get new perspectives.

I would like to suggest a new perspective by replacing Newton’s 1st law with two new laws:

1.1. A body moves in a series of random motions.
1.2. Observers see typical histories of a body when they measure outcomes in experiments.

These two new laws combined do not give Newton’s 1st law when put together. This is because Newton’s 1st law is a statistical phenomena that emerges from the conditions of the entire universe.

With other words, when an observer measures a force in an instant of time, both the force and the instant of time are statistical approximations of the series of random motions that the body goes through, that reflect the typical histories of that body.

There is no requirement that a body remains at rest, or in motion at a constant speed in a straight line, unless acted upon by a force. This fits the intuition of General Relativity, that a curved space-time can cause a body to accelerate in absence of forces acted upon it. In this case, the acceleration due to curved space-time is a statistical approximation of the series of random motions that the body goes through, that reflect the typical histories of that body. This means, no matter how much space-time is curved, a body might tunnel from one energy state to another in an instant.

It is also possible, by random chance, that the direction of time suddenly seemingly reverses, such that a body can start moving backwards. However, such events are unlikely, as they are not among the typical histories that observers see. The speed of a body is a statistical approximation. It is used to describe the direction of time, but a such direction of time is only an emergent phenomena.

This means, if, for some reason, all the observers in the entire universe suddenly start seeing a body move at half its previous speed, this would not be a violation of the laws of motion in principle. It would just be extremely unlikely to occur within the expected lifetime of a typical observer.

As a consequence, information of physical states might suddenly be created or destroyed. Preservation of information might be only a statical phenomena. However, it is also possible that there is another law of motion that preserves information over time, but it is independent of these 4 laws.

Now, the reason that Newton's 1st law approximately holds for observers like ourselves, is because the universe is rapidly expanding. The number of observers constantly increases, forming from local initial conditions that are similar to our local observable universe at the early moments it is possible to keep track of history. Since the universe is expanding rapidly, observers like ourselves will see typical histories of a body in motion at a constant speed in a straight line, unless it is acted upon by a force. It happens because a straight line is the shortest series of random events that can bring observers like ourselves to their current physical state, from the initial conditions.

One prediction of the two new laws of motion is that some particle might be formed from initial conditions of space-time, because this is the typical histories of some particle that observers see. The conditions to form the particle are present in the history of the local universe. Observers see typical histories of complex particles such as atoms in the table of elements, being formed through physical processes in stages that are relatively straight-forward from the initial conditions.

With other words, the typical histories will not contain many stages of complex formation of atoms that make the existence of observers like ourselves possible. This happens because as the universe expands, new observers like ourselves come into being, formed by similar initial conditions.

If there are many stages in the processes that form atoms needed for observers like ourselves, then such histories are less likely and hence not seen when doing experiments.

Imagine that some observer comes into being, by an incredibly rare random event, very early in the local universe. Since most observers do not come into being that way, a such history is unlikely. This is why observers like ourselves do not exist incredibly early. However, in the big picture, when considering the expected lifetime of the local universe, observers like ourselves exist relatively early. This means, there is are extremes, but they are not typical histories.

When observers see a body move in a straight line, it is because a such history is typical for observers like themselves. If observers see a body accelerate in some direction due to curved space-time, then it is because a such history is typical for observers like themselves. With other words, when observers see a body fall down with gravity, it is happens because observers like themselves see typical histories of the body falling down with gravity. They see it because they are located on a planet with a gravitational field. Which requires the planet to form before observers like themselves come into being. However, there are not many such stages of planet formation in the past, because most observers like themselves are formed from initial conditions that are similar, where the formation of the planet is a typical history, but relatively straight-forward from initial conditions to the current physical state.

As a consequence, observers like ourselves live on planets that are formed relatively early in time. Most planets that might contain life, are not formed yet. They will be at some point in the future.

In quantum physics, De Broglie-Bohm theory is a non-local interpretation where particles have definite positions, guided by a wave-function. In the double-slit experiment, a particle moves through a single slit, but its trajectory depends on the wave-function and hence the state of the two slits. If information is recorded about which slit the particle goes through, then the wave-function changes and the inference pattern of two slits disappears. An experiment using weak measurements produces similar trajectories as predicted by De Broglie-Bohm theory. Another experiment which sends two particles going through each slit at the same time causes inconsistency between Standard Quantum Mechanics and De Broglie-Bohm theory and confirms the predictions made by Standard Quantum Mechanics. Therefore, it seems that typical histories look sometimes like definite trajectories, but in general these typical histories are inconsistent with actual definite trajectories.

This evidence suggests that particles have typical histories when observed in experiments that might look like Newton's 1st law under De Broglie-Bohm theory, but the trajectories are not fully deterministic, violating Newton's 1st law in some cases. Instead of moving along the trajectory with a well-defined velocity, the particle goes through a series of random events. The nature of these random events are currently unknown, but in typical histories they confirm to probabilistic predictions in Standard Quantum Mechanics.

Standard Quantum Mechanics does not rule out the possibility that particles have definite positions only, but that they can not have both definite positions and obey Newton's 1st law. With other words, if particles have definite positions but goes through a series of random events, then this does not contradict the predictions of Standard Quantum Mechanics. However, to explain how such random events lead to typical histories as observed in experiments, it is needed to introduce more observers in the universe, such that what an observer sees in an experiment, is a typical history. The problem is to explain how such observers come into being.

The Everett many-worlds interpretation is often thought of as the universe branching locally into new observers, such that typical histories are what observers see across Everett branches. However, the predictions made by the Everett interpretation are identical to Standard Quantum Mechanics. Therefore, there is no experiment that can distinguish the two, since the theory is identical. Now, this means that there is a different way of thinking about the Everett interpretation. Locally, particles might have definite positions going through a series of random events. The Everett branches correspond to observers across the universe who exist in similar physical states prior to the measurements. When they measure, they see a typical history for observers like themselves. Hence, no local branching, but a global branching due to rapid expansion of the universe.

The two new laws replacing Newton's 1st law enables compatibility between quantum physics and relativity, but also allows physicists to use both De Broglie-Bohm theory and the Everett interpretation as approximate approaches. Particle trajectories are not fully deterministic, nor are Everett branches fully realistic. These two approaches are good approximations in different senses.

One reason that particles seem to have deterministic trajectories in most cases, is that when observers conduct experiments of this kind, they are measuring and collecting information about the trajectories of particles. Since observers like themselves are likely to look for the same knowledge, it would not be strange that trajectories look deterministic, since such trajectories would be typical histories for observers like themselves. With other words, the information that observers are looking for, is part of what makes observers see as likely histories. Observers like themselves are conducting similar experiments and looking for the same results. Therefore, it makes sense that particle trajectories look deterministic in most cases, but when you check closer, they do not.

In the case of the double-slit experiment where two particles are sent through each slit at the same time, they obey the predictions made by Standard Quantum Mechanics, which shows that these particles have no deterministic trajectories. These are typical histories for such particles, because in this experiment, the observers are not collecting information about deterministic trajectories. In fact, the experiment itself prevents such information from being collected, given that the two new laws hold instead of Newton's 1st law. If the particles had deterministic trajectories, then they would follow the predictions of De Broglie-Bohm theory. However, since they do not, this shows that they go through a series of random events. Now, since observers like themselves are not looking at typical histories that seem deterministic, they get typical histories that are not deterministic instead. As you would expect, from observers like themselves doing similar experiments.

The wave-function under the two new laws is not local. When new Everett branches appear due to decoherence, this is a statistical phenomena of typical histories across the universe for observers conducting experiments under similar conditions. Once one observer sets up an experiment, this is part of the definition of similar conditions to observers like oneself. The typical histories that such observers see depends on the setup of the experiment. This does not imply that the wave-function is local and continuously splitting up into Everett branches. The wave-function is a statistical description of similar physical states to the one being measured, but it is not possible to describe the system more accurately, since the motion goes through a series of random events. Therefore, the probabilistic distribution given by the wave-function is still the best tool for making predictions.

When observing the motion of a body at macro scale, it seems to follow Newton's 1st law. This is also a prediction of the classical limit due to decoherence in quantum physics. However, to tell that quantum physics is more accurate, one needs to measure the motion more precisely. Ironically, when an observer does not measure the motion more precisely, the typical history will seem to follow Newton's 1st law. In experiments which carefully avoids capturing information that might deviate from Newton's 1st law, quantum physics coincide with Newtonian mechanics and hence would not be inaccurate to say that typical histories will seem to follow Newton's 1st law. As there is no way to tell the difference, it makes no sense to say that the two new laws implies that in some cases, Newtonian mechanics might be followed more in typical histories than quantum physics. A more accurate theory makes the same predictions under conditions where a simpler theory might be used. Therefore, ontologically, there is no way to tell whether the selection effect "erases" information such that Newton's 1st law might be more accurate realistic description in some cases. All one can say, is that observers see typical histories when measuring outcomes in experiments.

Experiments can show seemingly contradictory results. When testing for deterministic trajectories, experiments shows that there are no deterministic trajectories. When testing for evolution in quantum super-position by repeated weak measurements, the system appears to freeze in time, showing what seems to be a deterministic trajectory in that case. No experiments reveal entirely whether motion goes through a series of random events only, or whether it only has a typical history without any random events. When testing for law 1.1. while 1.2. is false, the experiment gives negative result. Similarly, when testing for law 1.2. while 1.1. is false, the experiment gives negative result. This means, either both are false or both are true. Since no experiment can test whether both are false, but only possible to test whether both are true, the only experiments that could fail, shows that the two new laws survive. Therefore, the two new laws can not be proven to be true, but might be proven to be false in some future experiment.